Distances Between One-Dimensional Probability Laws Based on Cumulative and Quantile Functions

Abstract

This note catalogues the principal distances, discrepancies, and rank functionals built from cumulative distribution functions or quantiles on the real line. It distinguishes four questions that are often conflated: whether the object is a genuine metric, whether it is finite, whether it metrizes weak convergence, and whether it is quantitatively comparable with a Wasserstein distance. The main examples are Wasserstein–Mallows quantile distances, Kolmogorov–Smirnov and Kuiper distances, the Lévy metric, bounded-ground Wasserstein and Ky Fan metrics, Cramér and Cramér–von Mises discrepancies, Anderson–Darling weights, integrated quantile and stop-loss metrics, Lorenz curves, and ROC/AUC functionals.

Contents

1 Setup and the four different questions

Let $\alpha, \beta \in \mathcal{P}(\mathbb{R})$. Their cumulative distribution functions (CDFs) and left-continuous generalized quantiles are

$$\mathcal{C}_\alpha(x) := \alpha((-\infty, x]), \qquad q_\alpha(r) := \mathcal{C}_\alpha^{-1}(r) := \inf\{x \in \mathbb{R} : \mathcal{C}_\alpha(x) \geq r\}, \quad 0 < r < 1. \tag{1}$$

The CDF uses the spatial coordinate x, whereas the quantile uses the mass coordinate r. This is the notation of the OT4ML manuscript. The distinction explains most of the different scaling and topological behavior below.

For a proposed discrepancy d, four logically separate properties matter.

(i) *Domain and finiteness:* identify the largest natural class on which $d(\alpha, \beta) < +\infty$.

(ii) *Metric property:* $d(\alpha, \beta) = 0$ only when $\alpha = \beta$, and the triangle inequality holds.

(iii) *Weak topology:* $d(\alpha_n, \alpha) \to 0$ if and only if $\alpha_n \rightharpoonup \alpha$, where $\rightharpoonup$ denotes convergence in law, or weak (narrow) convergence against bounded continuous test functions; this is also often called weak-star convergence of probability measures.

(iv) *Quantitative comparison:* inequalities of the form $d \leq C\,\mathcal{W}_p^a$ or $\mathcal{W}_p \leq Cd^b$ hold on a specified class of laws.

Two distances can generate the same topology without being Lipschitz equivalent. Conversely, a popular test statistic need not be a metric at all. General references on probability metrics and empirical-distribution statistics are [8, 15, 3].

The basic convergence criteria are

$$\begin{aligned} \alpha_n \rightharpoonup \alpha \quad &\iff \quad \mathcal{C}_{\alpha_n}(x) \to \mathcal{C}_\alpha(x) \text{ at every continuity point of } \mathcal{C}_\alpha \\ &\iff \quad q_{\alpha_n}(r) \to q_\alpha(r) \text{ for a.e. } r \in (0,1). \end{aligned} \tag{2}$$

The last equivalence may equivalently be stated at every continuity point of q_α. A monotone function has only countably many discontinuities.

2 Quantile distances: Wasserstein and its variants

2.1 The Wasserstein–Mallows family

The canonical L^p distance between quantiles is exactly the one-dimensional Wasserstein distance [18, 19].

Proposition 2.1 (Quantile representation of Wasserstein distance)**.** *For $1 \leq p < \infty$ and $\alpha, \beta \in \mathcal{P}_p(\mathbb{R})$,*

$$\mathcal{W}_p(\alpha, \beta)^p = \int_0^1 |q_\alpha(r) - q_\beta(r)|^p \, \mathrm{d}r. \tag{3}$$

Moreover,

$$\mathcal{W}_\infty(\alpha, \beta) = \operatorname*{ess\,sup}_{0<r<1} |q_\alpha(r) - q_\beta(r)|. \tag{4}$$

Consequently, for $p < \infty$,

$$\mathcal{W}_p(\alpha_n, \alpha) \to 0 \quad \iff \quad \alpha_n \rightharpoonup \alpha \text{ and } \int |x|^p \,\mathrm{d}\alpha_n(x) \to \int |x|^p \,\mathrm{d}\alpha(x). \tag{5}$$

On a common compact interval, $\mathcal{W}_p$ therefore metrizes weak convergence.

Thus $\mathcal{W}_p$ does not metrize weak convergence on all of $\mathcal{P}(\mathbb{R})$: it also controls a moment. For example,

$$\beta_{\epsilon,R} := (1-\epsilon)\delta_0 + \epsilon\delta_R \rightharpoonup \delta_0 \quad \text{as } \epsilon \to 0, \qquad \mathcal{W}_p(\beta_{\epsilon,R}, \delta_0) = \epsilon^{1/p} R, \tag{6}$$

which need not tend to zero when $R \to \infty$. The $p = \infty$ topology is stronger still. Even on $[0,1]$, $\beta_n = (1 - 1/n)\delta_0 + n^{-1}\delta_1 \rightharpoonup \delta_0$ while $\mathcal{W}_\infty(\beta_n, \delta_0) = 1$.

2.2 A bounded quantile metric for weak convergence

Formula (2) gives a simple quantile metric that does metrize weak convergence without a moment condition.

Proposition 2.2 (Bounded quantile metric)**.** *The formula*

$$d_{\mathrm{q,bd}}(\alpha, \beta) := \int_0^1 \min\{1, |q_\alpha(r) - q_\beta(r)|\} \,\mathrm{d}r \tag{7}$$

defines a metric on $\mathcal{P}(\mathbb{R})$ and

$$d_{\mathrm{q,bd}}(\alpha_n, \alpha) \to 0 \quad \iff \quad \alpha_n \rightharpoonup \alpha.$$

Proof. The integrand is the bounded metric $\min\{1, |x - y|\}$, so the triangle inequality holds pointwise. Equality to zero identifies the two quantile functions almost everywhere and hence the two laws. Weak convergence gives almost-everywhere convergence of quantiles by (2); dominated convergence then proves one implication. Conversely, convergence in (7) is convergence in measure of the quantile functions. Every subsequence has an almost-everywhere convergent subsubsequence, which yields weak convergence by (2). □

This is best viewed as a canonical metric for convergence in measure of quantile functions. Unlike (3), it is not being asserted to be an optimal-transport cost for the truncated ground cost: a concave or truncated ground cost need not be minimized by the monotone coupling.

The less frequently used Ky Fan metric gives an equivalent quantile description of the same convergence. Applied to the canonical quantile coupling, it reads [3]

$$d_{\mathrm{q,KF}}(\alpha, \beta) := \inf\left\{\eta > 0 : \mathrm{Leb}\left\{r \in (0,1) : |q_\alpha(r) - q_\beta(r)| > \eta\right\} \le \eta\right\}. \tag{8}$$

It is a metric because the exceptional sets in two successive comparisons can be united. It metrizes weak convergence and satisfies

$$d_{\mathrm{q,bd}} \le 2 d_{\mathrm{q,KF}}, \qquad d_{\mathrm{q,KF}} \le d_{\mathrm{q,bd}}^{1/2}. \tag{9}$$

The first estimate splits the integral according to whether the quantile gap exceeds $d_{\mathrm{q,KF}}$; the second is Markov's inequality. The numerical scale matters in (8), because its threshold is used both as a spatial tolerance and as a probability tolerance.

2.3 Bounded-ground Wasserstein and Fortet–Mourier metrics

There is a related genuine transport metric that must not be confused with the fixed quantile coupling above. Equip $\mathbb{R}$ with the bounded metric $d_{\mathrm{b}}(x,y) := \min\{1,|x-y|\}$ and define

$$\mathcal{W}_{1,\mathrm{b}}(\alpha,\beta) := \inf_{\pi\in\mathcal{U}(\alpha,\beta)} \int_{\mathbb{R}^2} d_{\mathrm{b}}(x,y)\,\mathrm{d}\pi(x,y). \tag{10}$$

Kantorovich–Rubinstein duality identifies this, up to harmless normalization conventions, with the bounded-Lipschitz (or Dudley) and Fortet–Mourier metrics [6, 5, 8]. It metrizes weak convergence on $\mathcal{P}(\mathbb{R})$ and

$$\mathcal{W}_{1,\mathrm{b}}(\alpha,\beta) \le d_{\mathrm{q,bd}}(\alpha,\beta), \qquad \mathcal{W}_{1,\mathrm{b}}(\alpha,\beta) \le \min\{1,\mathcal{W}_1(\alpha,\beta)\}. \tag{11}$$

The first bound simply tests (10) with the quantile coupling. Equality need not hold because the truncated ground cost is not convex in the displacement.

For comparison, the following coupling form of the Prokhorov metric is due to Strassen [16, 3]:

$$d_{\mathrm{P}}(\alpha,\beta) := \inf\left\{\eta>0:\ \exists\pi\in\mathcal{U}(\alpha,\beta),\ \pi\{|x-y|>\eta\}\le\eta\right\}. \tag{12}$$

It also metrizes weak convergence, and direct coupling arguments give

$$\mathcal{W}_{1,\mathrm{b}} \le 2d_{\mathrm{P}}, \qquad d_{\mathrm{P}} \le \mathcal{W}_{1,\mathrm{b}}^{1/2}, \qquad d_{\mathrm{P}} \le d_{\mathrm{q,KF}}. \tag{13}$$

The last inequality reflects the fact that Prokhorov optimizes over all couplings, whereas (8) fixes the monotone one.

2.4 Weighted and trimmed quantile distances

For a measurable weight $w:(0,1)\to[0,\infty)$, a weighted quantile distance is

$$d_{\mathrm{q},p,w}(\alpha,\beta) := \left(\int_0^1 |q_\alpha(r)-q_\beta(r)|^p w(r)\,\mathrm{d}r\right)^{1/p}. \tag{14}$$

Weights large near 0 or 1 emphasize tails; weights that vanish there produce robust or trimmed comparisons. If $0<m\le w(r)\le M<\infty$ almost everywhere, then

$$m^{1/p}\,\mathcal{W}_p \le d_{\mathrm{q},p,w} \le M^{1/p}\,\mathcal{W}_p, \tag{15}$$

so the two metrics are bi-Lipschitz equivalent. If w vanishes on a set of positive measure, separation may fail. An unbounded weight near an endpoint requires additional tail integrability. Its topology is stronger than the $\mathcal{W}_p$ topology if w also has a positive lower bound; without that lower bound the two topologies can instead be incomparable.

A common robust version is the central trimmed distance

$$d_{\mathrm{q},p,\tau}^{\mathrm{trim}}(\alpha,\beta) := \left(\int_\tau^{1-\tau} |q_\alpha(r)-q_\beta(r)|^p\,\mathrm{d}r\right)^{1/p}, \qquad 0<\tau<\tfrac{1}{2}. \tag{16}$$

It is only a pseudometric: two laws can have identical central quantiles and different tails. It therefore does not metrize weak convergence on $\mathcal{P}(\mathbb{R})$. Optimally trimmed Wasserstein distances, in which a prescribed fraction of mass may be discarded before transport, are related but distinct robust-transport constructions.

2.5 Fractional and transformed quantile metrics

Two secondary constructions are useful when ordinary moments are not the appropriate scale. First, for $0 < s < 1$,

$$d_{\mathrm{q},s}(\alpha, \beta) := \int_0^1 |q_\alpha(r) - q_\beta(r)|^s \, \mathrm{d}r \tag{17}$$

is a metric on laws with finite s-th absolute moment: the triangle inequality follows from $|x - z|^s \le |x - y|^s + |y - z|^s$. Its convergence is equivalent to weak convergence plus convergence of the s-th absolute moments. Indeed, convergence of the metric gives quantile convergence in measure and $\big||x|^s - |y|^s\big| \le |x - y|^s$; the converse follows from almost-everywhere quantile convergence and uniform integrability. Despite its appearance, this is not denoted $\mathcal{W}_s$ here. For a concave cost $|x - y|^s$, the monotone coupling need not minimize the transport cost, and the OT4ML convention reserves $\mathcal{W}_p$ for $p \ge 1$.

Second, let $\psi : \mathbb{R} \to I$ be a strictly increasing homeomorphism onto an interval. The transformed-quantile metric

$$d_{p,\psi}(\alpha, \beta) := \left(\int_0^1 |\psi(q_\alpha(r)) - \psi(q_\beta(r))|^p \, \mathrm{d}r \right)^{1/p} = \mathcal{W}_p(\psi_\sharp \alpha, \psi_\sharp \beta) \tag{18}$$

is finite whenever the transformed p-moments are finite. A bounded choice, such as $\psi(x) = \arctan x$, metrizes weak convergence globally. On a common compact interval where $0 < m \le \psi' \le M$, it is bi-Lipschitz equivalent to $\mathcal{W}_p$, with constants m and M. On $\mathbb{R}_+$, the choice $\psi(x) = \log x$ instead measures multiplicative displacement and requires a finite p-th logarithmic moment; it is not equivalent to ordinary $\mathcal{W}_p$ without bounds away from zero and infinity.

3 CDF norm distances

3.1 The L^q-CDF family

For $1 \le q < \infty$, define

$$D_q(\alpha, \beta) := \left(\int_{\mathbb{R}} |\mathcal{C}_\alpha(x) - \mathcal{C}_\beta(x)|^q \, \mathrm{d}x \right)^{1/q}, \tag{19}$$

whenever finite, and define

$$D_\infty(\alpha, \beta) := \sup_{x \in \mathbb{R}} |\mathcal{C}_\alpha(x) - \mathcal{C}_\beta(x)|. \tag{20}$$

The special cases have different classical names:

- $D_1 = \mathcal{W}_1$ is the area between the two CDFs;
- D_2 is the Cramér distance;
- D_∞ is the Kolmogorov, uniform, or Kolmogorov–Smirnov distance.

Only $q = 1$ is simultaneously an L^q distance between CDFs and the same-order L^q distance between quantiles.

Proposition 3.1 (Topology of finite L^q-CDF distances)**.** *Let $1 \le q < \infty$.*

(i) *D_q is a finite metric on $\mathcal{P}_1(\mathbb{R})$.*

(ii) *$D_q(\alpha_n, \alpha) \to 0$ implies $\alpha_n \rightharpoonup \alpha$.*

(iii) *Weak convergence alone does not imply D_q convergence on the line.*

(iv) *On laws supported in a common compact interval, D_q metrizes weak convergence.*

For $q > 1$, the resulting topology on $\mathcal{P}_1(\mathbb{R})$ is strictly weaker than the $\mathcal{W}_1$ topology and strictly stronger than weak convergence in the sense of convergent sequences.

Proof. Since $|\mathcal{C}_\alpha - \mathcal{C}_\beta| \leq 1$ and the one-dimensional identity gives $\int |\mathcal{C}_\alpha - \mathcal{C}_\beta| = \mathcal{W}_1(\alpha, \beta)$, D_q is finite on $\mathcal{P}_1(\mathbb{R})$. The norm axioms give the triangle inequality, and equality almost everywhere of two right-continuous CDFs implies equality everywhere.

Suppose $D_q(\alpha_n, \alpha) \to 0$ and let x be a continuity point of $\mathcal{C}_\alpha$. If, along a subsequence, $\mathcal{C}_{\alpha_n}(x) \geq \mathcal{C}_\alpha(x) + \eta$, right-continuity of $\mathcal{C}_\alpha$ and monotonicity of $\mathcal{C}_{\alpha_n}$ produce an interval to the right of x on which the CDF difference is at least $\eta/2$. This contradicts L^q convergence. An interval to the left handles the opposite inequality. Thus $\mathcal{C}_{\alpha_n}(x) \to \mathcal{C}_\alpha(x)$, proving weak convergence.

For the failure of the converse, use (6) with $\epsilon_n = 1/n$ and $R_n = n^{q+1}$. Then $\beta_{\epsilon_n, R_n} \rightharpoonup \delta_0$, whereas $D_q^q = \epsilon_n^q R_n = n$. On a fixed compact interval, weak convergence gives pointwise convergence of CDFs outside a countable set; dominated convergence gives $D_q \to 0$.

Finally, for $q > 1$, choosing $R_n = n$ in the outlier example gives $D_q^q = n^{1-q} \to 0$ but $\mathcal{W}_1 = 1$. The previous counterexample shows that weak convergence need not imply D_q convergence. $\square$

3.2 Exact comparison with Wasserstein distances

The CDF family admits simple sharp-scale comparisons.

Proposition 3.2 (CDF norms versus Wasserstein distances)**.** *Let $1 \leq q < \infty$ and $1 \leq p < \infty$. Whenever the right-hand side is finite,*

$$D_q(\alpha, \beta)^q \leq \mathcal{W}_1(\alpha, \beta) \leq \mathcal{W}_p(\alpha, \beta). \tag{21}$$

If both laws are supported in an interval of length L, then

$$\mathcal{W}_p(\alpha, \beta) \leq L^{1-1/(pq)} D_q(\alpha, \beta)^{1/p}. \tag{22}$$

For $q = \infty$, compact support gives

$$\mathcal{W}_p(\alpha, \beta) \leq L D_\infty(\alpha, \beta)^{1/p}. \tag{23}$$

Proof. The first inequality follows from $|\mathcal{C}_\alpha - \mathcal{C}_\beta|^q \leq |\mathcal{C}_\alpha - \mathcal{C}_\beta|$ and the identity $D_1 = \mathcal{W}_1$. On an interval of length L, Hölder's inequality and bounded displacement give

$$\mathcal{W}_1 \leq L^{1-1/q} D_q, \qquad \mathcal{W}_p^p \leq L^{p-1} \mathcal{W}_1 .$$

Combining them proves (22). The same argument with $\mathcal{W}_1 \leq L D_\infty$ proves (23). $\square$

These are Hölder, not generally Lipschitz, comparisons. For the translated Diracs $\alpha = \delta_0$ and $\beta = \delta_h$,

$$\mathcal{W}_p(\delta_0, \delta_h) = |h|, \qquad D_q(\delta_0, \delta_h) = |h|^{1/q}, \qquad D_\infty(\delta_0, \delta_h) = 1 \quad (h \neq 0). \tag{24}$$

Thus D_q and $\mathcal{W}_p$ are not generally Lipschitz equivalent when $q > 1$, and Kolmogorov distance is not even continuous with respect to $\mathcal{W}_p$ at an atomic law.

3.3 Cramér distance, energy distance, and CRPS

The unweighted L^2-CDF distance should be distinguished from the distribution-weighted Cramér–von Mises statistic discussed below. If $X, X' \sim \alpha$ and $Y, Y' \sim \beta$ are independent, then

$$2D_2(\alpha,\beta)^2 = 2\mathbb{E}|X-Y| - \mathbb{E}|X-X'| - \mathbb{E}|Y-Y'|. \tag{25}$$

Under the metric convention in which the right-hand side is the *squared* energy distance, the energy distance is therefore $\sqrt{2}D_2$; some statistical sources call the unrooted expression itself the energy distance [17]. Hence Cramér distance is a genuine metric on $\mathcal{P}_1(\mathbb{R})$, but Proposition 3.1 shows that it does not metrize weak convergence on the whole line and is strictly weaker than $\mathcal{W}_1$ in its tail control.

The continuous ranked probability score (CRPS) is the proper scoring rule

$$\mathrm{CRPS}(\alpha, y) \coloneqq \int_{\mathbb{R}} \left(\mathcal{C}_\alpha(x) - \mathbf{1}_{\{y \le x\}}\right)^2 \mathrm{d}x.$$

If $Y \sim \beta$, subtracting the expected score of the true CDF gives

$$\mathbb{E}\,\mathrm{CRPS}(\alpha, Y) - \mathbb{E}\,\mathrm{CRPS}(\beta, Y) = D_2(\alpha,\beta)^2. \tag{26}$$

Thus CRPS is not another pairwise distance: its excess population risk is the squared Cramér distance.

4 Supremum CDF distances

4.1 Kolmogorov–Smirnov and Kuiper

The Kolmogorov–Smirnov distance is D_∞ from (20). The one-sided statistics

$$D^+(\alpha,\beta) \coloneqq \sup_x(\mathcal{C}_\alpha(x) - \mathcal{C}_\beta(x)), \qquad D^-(\alpha,\beta) \coloneqq \sup_x(\mathcal{C}_\beta(x) - \mathcal{C}_\alpha(x))$$

are not symmetric metrics individually. Their maximum is the KS distance. Their sum is the Kuiper distance

$$d_{\mathrm{Kp}}(\alpha,\beta) \coloneqq D^+(\alpha,\beta) + D^-(\alpha,\beta), \qquad D_\infty \le d_{\mathrm{Kp}} \le 2D_\infty. \tag{27}$$

Kuiper's statistic is especially natural on the circle because the sum is insensitive to the choice of the cut point [10]. On the line it has exactly the same convergence properties as KS.

There is an equivalent discrepancy interpretation. If rays $(-\infty, x]$ in the KS definition are replaced by bounded half-open intervals, then

$$\begin{aligned} d_{\mathrm{int}}(\alpha,\beta) &\coloneqq \sup_{a<b} |\alpha((a,b]) - \beta((a,b])| \\ &= \sup_{a<b} |(\mathcal{C}_\alpha - \mathcal{C}_\beta)(b) - (\mathcal{C}_\alpha - \mathcal{C}_\beta)(a)| = d_{\mathrm{Kp}}(\alpha,\beta). \end{aligned} \tag{28}$$

Indeed, $\mathcal{C}_\alpha - \mathcal{C}_\beta$ vanishes at both ends of the line, so its range is $D^+ + D^-$. Thus KS is the ray, or star-discrepancy, metric and Kuiper is the interval-discrepancy metric. Both are bounded above by total variation, which takes the supremum over all Borel sets. Total variation is much stronger and does not metrize weak convergence: it remains one between $\delta_{1/n}$ and δ_0.

Proposition 4.1 (Kolmogorov topology and regularity comparison)**.** *Kolmogorov convergence implies weak convergence. Conversely, if $\alpha_n \rightharpoonup \alpha$ and $\mathcal{C}_\alpha$ is continuous, then $D_\infty(\alpha_n, \alpha) \to 0$. Hence KS metrizes weak convergence at every atomless limiting law, but not on all of $\mathcal{P}(\mathbb{R})$.*

If α and β have densities bounded by M and $p \ge 1$, then

$$D_\infty(\alpha,\beta) \le (p+1)p^{-p/(p+1)} M^{p/(p+1)}\, \mathcal{W}_p(\alpha,\beta)^{p/(p+1)}. \tag{29}$$

Proof. The first statement is Pólya's uniform-convergence theorem for distribution functions; the counterexample at an atom is $\delta_{1/n} \rightharpoonup \delta_0$ with KS distance one.

For the quantitative bound, couple $X \sim \alpha$ and $Y \sim \beta$. For $h > 0$, monotonicity and the density bound give

$$\mathcal{C}_\alpha(x) - \mathcal{C}_\beta(x) \le \mathbb{P}(|X - Y| > h) + \mathcal{C}_\beta(x + h) - \mathcal{C}_\beta(x) \le \frac{\mathbb{E}|X - Y|^p}{h^p} + Mh.$$

Interchanging the laws controls the opposite difference. Take an optimal coupling and minimize $\mathcal{W}_p^p h^{-p} + Mh$ over $h > 0$. □

The density assumption is essential: the translated-Dirac example (24) rules out every bound $D_\infty \le C\,\mathcal{W}_p^a$ with $a > 0$ on arbitrary laws. Conversely, (23) shows that KS controls $\mathcal{W}_p$ in Hölder form on a common compact support.

4.2 The Lévy metric

The Lévy metric softens both the horizontal and vertical comparison of CDFs:

$$d_{\mathrm{L}}(\alpha, \beta) \coloneqq \inf\{\epsilon > 0 : \mathcal{C}_\alpha(x - \epsilon) - \epsilon \le \mathcal{C}_\beta(x) \le \mathcal{C}_\alpha(x + \epsilon) + \epsilon \text{ for every } x\}. \tag{30}$$

Unlike KS, it permits a small horizontal displacement of an atom.

Proposition 4.2 (Lévy metric and weak convergence)**.** *The Lévy metric metrizes weak convergence on all of $\mathcal{P}(\mathbb{R})$. Moreover,*

$$d_{\mathrm{L}}(\alpha, \beta) \le D_\infty(\alpha, \beta), \qquad d_{\mathrm{L}}(\alpha, \beta) \le \min\{1, \mathcal{W}_p(\alpha, \beta)^{p/(p+1)}\}. \tag{31}$$

There is no converse control of $\mathcal{W}_p$ by d_{L} on an unbounded space without a moment or tail assumption.

Proof. The metrization statement is classical [11, 3]; the first inequality is immediate from the definitions. For the second, an optimal p-transport coupling and Markov's inequality give

$$\mathbb{P}(|X - Y| > \epsilon) \le \mathcal{W}_p^p / \epsilon^p.$$

Taking $\epsilon = \mathcal{W}_p^{p/(p+1)}$ and using the coupling characterization of the Prokhorov metric yields $d_{\mathrm{L}} \le d_{\mathrm{P}} \le \epsilon$. Finally, the outlier family (6) can converge weakly, and hence in Lévy metric, while its p-Wasserstein distance stays bounded away from zero or diverges. □

On a common compact interval, Lévy and every finite $\mathcal{W}_p$ generate the same topology, because both metrize weak convergence there. This topological equivalence should not be mistaken for a scale-free global Lipschitz equivalence.

5 Weighted CDF discrepancies and goodness-of-fit statistics

5.1 Fixed weighted CDF metrics

Let ν be a finite Borel measure on $\mathbb{R}$. The fixed-reference weighted family is

$$D_{q,\nu}(\alpha, \beta) \coloneqq \left(\int_{\mathbb{R}} |\mathcal{C}_\alpha - \mathcal{C}_\beta|^q \,\mathrm{d}\nu \right)^{1/q}. \tag{32}$$

Replacing Lebesgue measure by a finite reference removes the tail obstruction of Proposition 3.1.

Proposition 5.1 (A weighted CDF metric for weak convergence). *Assume that ν is finite, nonatomic, and gives positive mass to every nonempty open interval. Then $D_{q,\nu}$ is a metric on $\mathcal{P}(\mathbb{R})$ and metrizes weak convergence for every $1 \le q < \infty$.*

Proof. Full support and right-continuity ensure separation. If $\alpha_n \rightharpoonup \alpha$, the CDFs converge outside the countable set of discontinuities of $\mathcal{C}_\alpha$. This set is ν-null, so dominated convergence applies. Conversely, the monotonicity argument from Proposition 3.1 works because every interval has positive ν-mass. □

For example, one may take ν to be a Gaussian or logistic law. This metric depends on the chosen coordinate system and reference measure. If $\mathrm{d}\nu = w(x)\,\mathrm{d}x$ with $\|w\|_\infty < \infty$, then

$$D_{q,\nu}^q \le \|w\|_\infty \, \mathcal{W}_1 \,. \tag{33}$$

There is generally no reverse moment control: a reference with rapidly decaying tails deliberately makes remote discrepancies inexpensive.

5.2 Cramér–von Mises, Anderson–Darling, and Watson

Classical goodness-of-fit statistics usually integrate against the reference distribution rather than against Lebesgue measure. Taking β as the target, the population Cramér–von Mises (CvM) discrepancy is

$$\mathrm{CvM}_\beta(\alpha)^2 := \int_{\mathbb{R}} (\mathcal{C}_\alpha(x) - \mathcal{C}_\beta(x))^2 \,\mathrm{d}\beta(x). \tag{34}$$

If $\mathcal{C}_\beta$ is continuous, then

$$\mathrm{CvM}_\beta(\alpha)^2 = \int_0^1 \big(\mathcal{C}_\alpha(q_\beta(r)) - r\big)^2 \,\mathrm{d}r. \tag{35}$$

The expression is the L^2 deviation of the probability–probability (P–P), or ordinal-dominance, curve from the diagonal. It is distribution-free under the null after the probability integral transform [15]. This is not the Cramér distance D_2, which uses $\mathrm{d}x$ and is tied to energy distance by (25).

The Anderson–Darling discrepancy gives more weight to both tails:

$$\mathrm{AD}_\beta(\alpha)^2 := \int_{\mathbb{R}} \frac{(\mathcal{C}_\alpha(x) - \mathcal{C}_\beta(x))^2}{\mathcal{C}_\beta(x)(1-\mathcal{C}_\beta(x))} \,\mathrm{d}\beta(x) = \int_0^1 \frac{|\mathcal{C}_\alpha(q_\beta(r)) - r|^2}{r(1-r)} \,\mathrm{d}r, \tag{36}$$

when the expressions are finite [1]. Watson's U^2 statistic is a centered CvM statistic used for circular data; it removes the mean vertical displacement of the P–P curve so that the statistic is insensitive to the arbitrary circular origin.

These objects require care when called "distances." The one-sample forms (34)–(36) are asymmetric because β is privileged. They can fail to separate laws when β does not have sufficiently rich support, and the Anderson–Darling value can be infinite. Two-sample versions often replace β by the pooled law $(\alpha + \beta)/2$; these are symmetric test statistics, but their pair-dependent integration measure does not automatically yield a triangle inequality. They also do not metrize weak convergence globally: $\delta_{1/n} \rightharpoonup \delta_0$ is again a counterexample for statistics that inspect the jump of the atomic reference. At a continuous limiting CDF, dominated versions behave much more like KS and CvM empirical-process statistics.

6 Integrated CDFs and integrated quantiles

6.1 Stop-loss metric

For $\alpha \in \mathcal{P}_1(\mathbb{R})$, its stop-loss, call, or integrated-survival transform is

$$S_\alpha(t) := \int_{\mathbb{R}} (x-t)_+ \,\mathrm{d}\alpha(x) = \int_t^\infty (1 - \mathcal{C}_\alpha(x))\,\mathrm{d}x. \tag{37}$$

It is a convex function whose one-sided derivatives recover the survival function, so it determines the law. The stop-loss metric is

$$d_{\mathrm{SL}}(\alpha, \beta) := \sup_{t \in \mathbb{R}} |S_\alpha(t) - S_\beta(t)|. \tag{38}$$

It is classical in stochastic-order and actuarial theory [14, 4].

Proposition 6.1 (Stop-loss and $\mathcal{W}_1$ topologies)**.** *The stop-loss formula defines a metric on $\mathcal{P}_1(\mathbb{R})$ and*

$$d_{\mathrm{SL}}(\alpha, \beta) \leq \mathcal{W}_1(\alpha, \beta). \tag{39}$$

Moreover, $d_{\mathrm{SL}}(\alpha_n, \alpha) \to 0$ if and only if $\mathcal{W}_1(\alpha_n, \alpha) \to 0$. Thus the two metrics induce the same topology, but they are not globally bi-Lipschitz equivalent.

Proof. Every function $x \mapsto (x - t)_+$ is 1-Lipschitz, so (39) follows from Kantorovich–Rubinstein duality. The transform determines its law through its one-sided derivatives, proving separation; the other metric axioms are immediate.

Suppose $d_{\mathrm{SL}}(\alpha_n, \alpha) \to 0$. Uniform convergence of the convex functions S_{α_n} implies convergence of their derivatives at every differentiability point of S_α, hence weak convergence of the laws. In addition, $S_{\alpha_n}(0) \to S_\alpha(0)$ gives convergence of the positive first moments. Finally, $S_\alpha(t) + t \to \int x \, \mathrm{d}\alpha(x)$ as $t \to -\infty$, and uniform convergence of the transforms gives convergence of the means. Positive and negative first moments therefore converge, which together with weak convergence is equivalent to $\mathcal{W}_1$ convergence. The converse follows from (39).

For failure of a reverse Lipschitz bound, put equal masses on the pairs $\{4k, 4k+3\}$ for α_n and $\{4k+1, 4k+2\}$ for β_n, $k = 0, \ldots, n-1$. Their CDF difference alternates in sign on each block, giving $\mathcal{W}_1(\alpha_n, \beta_n) = 1$ but $d_{\mathrm{SL}}(\alpha_n, \beta_n) = 1/(2n)$. □

The stop-loss metric is therefore a useful example of topological equivalence without Lipschitz equivalence. It is also a natural metric for convex-order questions because $\alpha \preceq_{\mathrm{cx}} \beta$ is characterized, under equality of means, by $S_\alpha(t) \leq S_\beta(t)$ for every t.

6.2 Integrated quantiles and Lorenz curves

The lower integrated quantile, or generalized Lorenz curve, is

$$M_\alpha(r) := \int_0^r q_\alpha(s) \, \mathrm{d}s, \qquad 0 \leq r \leq 1. \tag{40}$$

This is the notation already used for integrated quantiles in the OT4ML manuscript. It determines q_α by differentiation almost everywhere and hence determines α.

Proposition 6.2 (Integrated-quantile metric)**.** *The formula*

$$d_{\mathrm{IQ}}(\alpha, \beta) := \sup_{0 \leq r \leq 1} |M_\alpha(r) - M_\beta(r)| \tag{41}$$

defines a metric on $\mathcal{P}_1(\mathbb{R})$, and

$$d_{\mathrm{IQ}}(\alpha, \beta) \leq \int_0^1 |q_\alpha - q_\beta| \, \mathrm{d}r = \mathcal{W}_1(\alpha, \beta), \tag{42}$$

while $d_{\mathrm{IQ}}(\alpha_n, \alpha) \to 0$ if and only if $\mathcal{W}_1(\alpha_n, \alpha) \to 0$. The two metrics are not globally bi-Lipschitz equivalent.

Proof. Only the converse topological implication requires explanation. Uniform convergence of the convex functions M_{α_n} implies convergence of their derivatives at every differentiability point of M_α, hence almost-everywhere convergence of q_{α_n} to q_α and weak convergence of the laws. In addition,

$$\int x_- \,\mathrm{d}\alpha(x) = -\min_{0\le r\le 1} M_\alpha(r), \qquad \int x_+ \,\mathrm{d}\alpha(x) = M_\alpha(1) - \min_{0\le r\le 1} M_\alpha(r).$$

Uniform convergence therefore gives convergence of both first-moment tails, which upgrades weak convergence to $\mathcal{W}_1$ convergence. The alternating atomic example in the proof of Proposition 6.1 has $\mathcal{W}_1 = 1$ but $d_{\mathrm{IQ}} = 1/(2n)$, ruling out a reverse Lipschitz bound. $\square$

Integrated quantiles are convex conjugates of integrated CDF or stop-loss transforms, and are standard tools for convex and increasing-convex stochastic orders.

For a nonnegative law with mean $m_\alpha > 0$, the normalized Lorenz curve is

$$L_\alpha(r) := \frac{M_\alpha(r)}{m_\alpha}. \tag{43}$$

Distances such as $\|L_\alpha - L_\beta\|_1$ or $\|L_\alpha - L_\beta\|_\infty$ compare inequality profiles [12, 7]. They are not metrics on probability laws: $L_{(x\mapsto ax)_\sharp\alpha} = L_\alpha$ for every $a > 0$. They become metrics only after quotienting by positive scale or after retaining the mean as an additional coordinate. Consequently, a Lorenz-curve distance alone cannot metrize weak convergence or be equivalent to $\mathcal{W}_p$ on the original line.

The Gini coefficient is twice the area between one Lorenz curve and the diagonal. It is a scalar summary of one law, not a distance between two laws.

6.3 Higher integrated CDFs and Zolotarev metrics

Zolotarev's ideal metrics extend the Kantorovich–Rubinstein viewpoint by testing against functions with higher-order Hölder derivatives [20]. If $s = m + \gamma$, where $m \in \mathbb{N}_0$ and $0 < \gamma \le 1$, one considers

$$\zeta_s(\alpha,\beta) := \sup_f \left| \int f \,\mathrm{d}(\alpha-\beta) \right|, \qquad |f^{(m)}(x) - f^{(m)}(y)| \le |x-y|^\gamma. \tag{44}$$

For $s = 1$, this is $\mathcal{W}_1$. For $m \ge 1$, finiteness requires the moments through order m, which are invisible to the Hölder seminorm, to agree; finite s-th absolute moments are a standard sufficient integrability assumption. Repeated integration by parts then represents the metric through iterated primitives of $\mathcal{C}_\alpha - \mathcal{C}_\beta$, generalizing the stop-loss construction. These are moment-sensitive approximation metrics, not metrics for weak convergence on all of $\mathcal{P}(\mathbb{R})$.

7 P–P curves, ROC curves, and AUC

7.1 Ordinal-dominance and ROC representations

Assume first that $\mathcal{C}_{\alpha_0}$ is continuous and strictly increasing. The ordinal-dominance, or P–P, curve of α_1 relative to α_0 is

$$\mathrm{ODC}_{0,1}(u) := \mathcal{C}_{\alpha_1}(q_{\alpha_0}(u)), \qquad 0 < u < 1. \tag{45}$$

For score laws α_0 in the negative class and α_1 in the positive class, one common ROC convention is

$$\mathrm{ROC}_{0,1}(u) := 1 - \mathcal{C}_{\alpha_1}(q_{\alpha_0}(1-u)), \qquad 0 < u < 1, \tag{46}$$

where u is the false-positive rate. ROC is a reflected ODC curve [2, 9].

Several CDF discrepancies are simply norms of this curve's deviation from the diagonal:

$$D_\infty(\alpha_0, \alpha_1) = \sup_{0<u<1} |\mathrm{ODC}_{0,1}(u) - u|, \tag{47}$$

$$\mathrm{CvM}_{\alpha_0}(\alpha_1)^2 = \int_0^1 |\mathrm{ODC}_{0,1}(u) - u|^2 \,\mathrm{d}u, \tag{48}$$

$$\mathrm{AD}_{\alpha_0}(\alpha_1)^2 = \int_0^1 \frac{|\mathrm{ODC}_{0,1}(u) - u|^2}{u(1-u)} \,\mathrm{d}u. \tag{49}$$

Thus KS is the maximal vertical P–P deviation, CvM its average squared deviation, and Anderson–Darling a tail-weighted version.

These rank constructions are invariant when the same strictly increasing transformation is applied to both score distributions. Wasserstein distances are instead sensitive to the physical scale. Therefore no two-sided Lipschitz or Hölder equivalence with $\mathcal{W}_p$ can hold without fixing a scale and imposing regularity bounds.

7.2 Area under the ROC curve is not a probability metric

For independent $X_0 \sim \alpha_0$ and $X_1 \sim \alpha_1$, the tie-corrected area under the ROC curve is

$$\mathrm{AUC}(\alpha_0, \alpha_1) := \mathbb{P}(X_1 > X_0) + \tfrac{1}{2}\mathbb{P}(X_1 = X_0). \tag{50}$$

For continuous laws this is $\mathbb{P}(X_1 > X_0) = \int \mathcal{C}_{\alpha_0}(y)\,\mathrm{d}\alpha_1(y)$ and equals the population Mann–Whitney–Wilcoxon ranking functional [13, 2].

Neither AUC nor $|\mathrm{AUC} - 1/2|$ is a distance between laws. It is oriented, does not satisfy a triangle inequality, and fails separation. For example, if α_0 and α_1 are two different centered symmetric laws, then $X_1 - X_0$ is symmetric and $\mathrm{AUC} = 1/2$ (when ties have zero probability). Taking $\alpha_0 = \mathcal{N}(0,1)$ and $\alpha_1 = \mathcal{N}(0,4)$ gives distinct laws with zero AUC deviation from 1/2. AUC is also invariant under every common increasing reparameterization of scores, whereas $\mathcal{W}_p$ changes under nonlinear reparameterization and scales linearly under dilation.

For laws with finite first moments, three different "areas" must therefore not be confused:

(i) $\int |\mathcal{C}_\alpha - \mathcal{C}_\beta|\,\mathrm{d}x = \mathcal{W}_1(\alpha, \beta)$ is the absolute area between CDFs and is a metric;

(ii) $\int (\mathcal{C}_\alpha - \mathcal{C}_\beta)\,\mathrm{d}x = \int x\,\mathrm{d}\beta - \int x\,\mathrm{d}\alpha$ is signed area and only compares means;

(iii) area under an ROC curve is the ranking probability (50), not a transport distance.

8 A compact comparison catalogue

The following table summarizes the main objects. "Weak globally" means on all of $\mathcal{P}(\mathbb{R})$; moment-restricted statements are indicated separately.

Object	Formula	Metric status	Convergence and relation to Wasserstein
Wasserstein–Mallows $\mathcal{W}_p$, $p < \infty$	$\|q_\alpha - q_\beta\|_{L^p(0,1)}$	Metric on $\mathcal{P}_p(\mathbb{R})$	Metrizes weak convergence plus p-moment convergence; weak on common compact support.

Object	Formula	Metric status	Convergence and relation to Wasserstein
$\mathcal{W}_\infty$	$\lVert q_\alpha - q_\beta \rVert_\infty$	Extended metric	Strictly stronger than weak convergence, even on compact support.
Bounded quantile metric	$\int_0^1 \min(1, \lvert q_\alpha - q_\beta \rvert)$	Metric on $\mathcal{P}(\mathbb{R})$	Metrizes weak convergence globally.
Quantile Ky Fan	$\inf\{\eta : \mathrm{Leb}(\lvert q_\alpha - q_\beta \rvert > \eta) \le \eta\}$	Metric on $\mathcal{P}(\mathbb{R})$	Metrizes weak convergence; quantitatively equivalent to the bounded quantile metric as in (9).
Bounded-ground $\mathcal{W}_{1,\mathrm{b}}$	OT for $d_\mathrm{b} = \min(1, \lvert x - y \rvert)$	Metric on $\mathcal{P}(\mathbb{R})$	Fortet–Mourier type; metrizes weak convergence and is no larger than the bounded quantile metric.
Prokhorov	Best coupling with $\pi\{\lvert x - y \rvert > \eta\} \le \eta$	Metric on $\mathcal{P}(\mathbb{R})$	Metrizes weak convergence; related to $\mathcal{W}_{1,\mathrm{b}}$ by (13).
Weighted quantile $d_{\mathrm{q},p,w}$	$\left(\int \lvert q_\alpha - q_\beta \rvert^p w \right)^{1/p}$	Metric if $w > 0$ a.e.; possibly extended	Bi-Lipschitz to $\mathcal{W}_p$ if w is bounded above and below; otherwise changes tail sensitivity.
Trimmed quantile	$\left(\int_\tau^{1-\tau} \lvert q_\alpha - q_\beta \rvert^p \right)^{1/p}$	Pseudometric	Ignores tails; does not metrize weak convergence.
Fractional quantile $d_{\mathrm{q},s}$	$\int_0^1 \lvert q_\alpha - q_\beta \rvert^s$, $0 < s < 1$	Metric under finite s-moments	Metrizes weak convergence plus s-moment convergence; it is not the concave-cost OT value.
Transformed quantile $d_{p,\psi}$	$\lVert \psi \circ q_\alpha - \psi \circ q_\beta \rVert_{L^p}$	Metric under transformed moments	A bounded homeomorphism ψ gives the weak topology; local derivative bounds give comparison with $\mathcal{W}_p$.
L^q CDF, $q < \infty$	$D_q = \lVert \mathcal{C}_\alpha - \mathcal{C}_\beta \rVert_{L^q(\,\mathrm{d}x)}$	Metric on $\mathcal{P}_1(\mathbb{R})$	$D_q^q \le \mathcal{W}_1$; implies weak convergence, but weak convergence does not imply D_q convergence without tail control.
Cramér / energy	D_2; squared energy $= 2D_2^2$	Metric on $\mathcal{P}_1(\mathbb{R})$	Strictly weaker in tails than $\mathcal{W}_1$; weak on common compact support.

Object	Formula	Metric status	Convergence and relation to Wasserstein
Kolmogorov–Smirnov	$\|\mathcal{C}_\alpha - \mathcal{C}_\beta\|_\infty$	Metric	Implies weak; converse at continuous limiting CDFs, but not at atoms. No control by $\mathcal{W}_p$ without regularity.
Kuiper	$\sup(F-G)+\sup(G-F)$	Metric; equivalent to KS	Same topology as KS; equals interval discrepancy and is adapted to circular data.
Lévy	Horizontal and vertical CDF tolerance	Metric on $\mathcal{P}(\mathbb{R})$	Metrizes weak convergence globally; $d_{\mathrm{L}} \lesssim \mathcal{W}_p^{p/(p+1)}$, but no reverse tail control.
Fixed weighted CDF	$\|\mathcal{C}_\alpha - \mathcal{C}_\beta\|_{L^q(\nu)}$	Metric if ν has full support	Metrizes weak convergence if ν is finite, nonatomic, and has full support.
CvM / Anderson–Darling	$\int (F-G)^2 \, \mathrm{d}G$; tail-weighted analogue	Usually asymmetric discrepancies, not metrics	Rank-based test statistics; no global weak metrization in their usual reference-dependent form.
Stop-loss	$\sup_t \lvert \mathbb{E}(X-t)_+ - \mathbb{E}(Y-t)_+ \rvert$	Metric on $\mathcal{P}_1(\mathbb{R})$	Same topology as $\mathcal{W}_1$ and bounded above by $\mathcal{W}_1$, but not bi-Lipschitz equivalent.
Integrated quantile	$\sup_{r\in[0,1]} \lvert M_\alpha(r) - M_\beta(r) \rvert$	Metric on $\mathcal{P}_1(\mathbb{R})$	Same topology as $\mathcal{W}_1$ and bounded above by $\mathcal{W}_1$, but not bi-Lipschitz equivalent.
Zolotarev ζ_s	Smooth test functions; equivalently iterated CDF primitives	Metric on moment-matched classes	$\zeta_1 = \mathcal{W}_1$; higher orders impose moment matching and do not metrize weak convergence globally.
Lorenz-curve norm	$\|L_\alpha - L_\beta\|$	Pseudometric on laws; metric modulo scale	Scale invariant, hence cannot metrize weak convergence of unnormalized laws or be equivalent to $\mathcal{W}_p$.
ROC/AUC	$\mathbb{P}(X_1 > X_0) + \frac{1}{2}\mathbb{P}(X_1 = X_0)$	Ranking functional, not a metric	Monotone-transform invariant and non-separating; no Wasserstein equivalence and no induced topology on laws.

9 Two counterexamples that diagnose most claims

The following two families are useful consistency checks for any proposed comparison.

Small translation. For δ_0 and δ_h, formula (24) gives $\mathcal{W}_p = |h|$, $D_q = |h|^{1/q}$, and $\mathrm{KS} = 1$. Thus spatial transport metrics see the size of the displacement, finite CDF norms see both displacement and the selected exponent, and rank suprema see only that a jump moved.

Rare remote outlier. For δ_0 and $\beta_{\epsilon,R}$ from (6),

$$\mathcal{W}_p = \epsilon^{1/p}R, \qquad D_q = \epsilon R^{1/q}, \qquad D_\infty = \epsilon, \qquad d_{\mathrm{L}} \to 0 \quad \text{when } \epsilon \to 0. \tag{51}$$

The free choices of ϵ and R show exactly which quantities control tail location, tail mass, or only weak convergence.

10 Practical reading guide

- Use $\mathcal{W}_p$ when the geometry and the size of displacement matter and a p-moment assumption is meaningful.
- Use Lévy, Prokhorov, or the bounded quantile metric (7) when the sole target is convergence in law. The bounded-ground Wasserstein metric is the OT/Fortet–Mourier version; the Ky Fan formula is useful when probability and spatial tolerances should be shown separately.
- Use KS or Kuiper for uniform rank discrepancies, remembering that atoms make them strictly stronger than weak convergence.
- Use Cramér/energy distance or CRPS when an integrated squared CDF error is desired; do not confuse it with the reference-weighted CvM statistic.
- Use CvM or Anderson–Darling for goodness-of-fit testing after a probability-integral transform; they are not generic transport metrics.
- Use weighted or trimmed quantile norms when tail emphasis or robustness is intentional, and state the weight because it determines the topology. Transformed quantiles are preferable when displacement is naturally measured after a monotone change of scale, such as a logarithm.
- Use stop-loss or integrated quantiles for convex-order and actuarial questions. Use normalized Lorenz curves only when scale invariance is intended.
- Treat ROC and AUC as two-sample ranking summaries, not as distances on the Wasserstein space.

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